
SMART VISIT

From 5,100 places to one coherent itinerary

Personalized tourist itinerary recommendation
based on user preferences and contextual data

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Master D2SM · ENSAM Meknes · 2025-2026

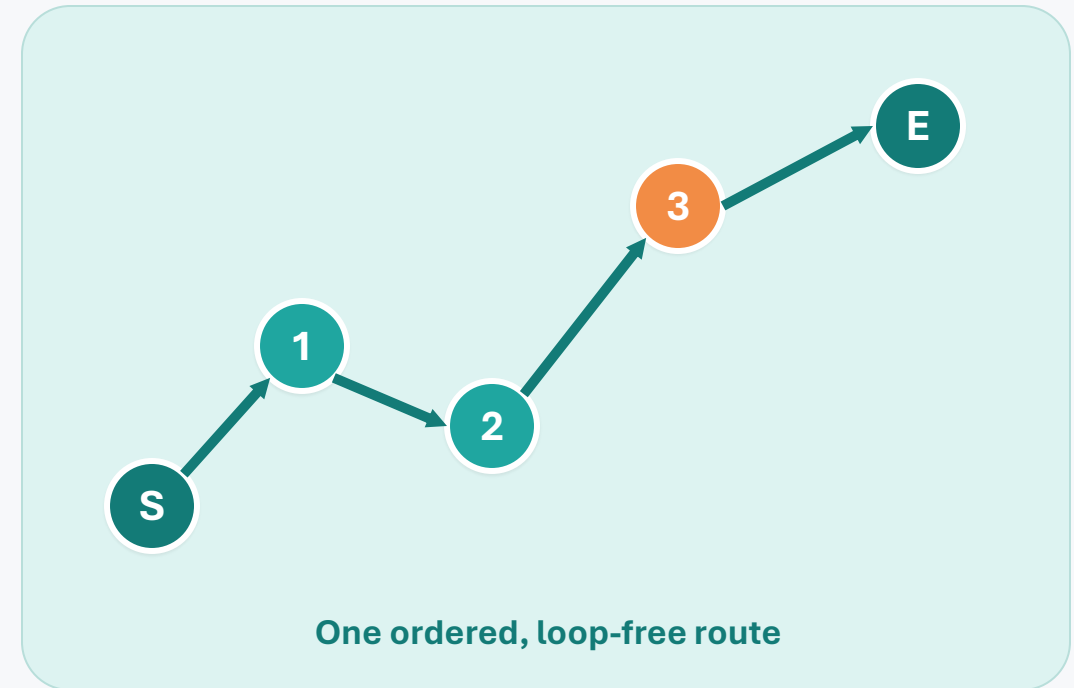


A good trip answers two questions: what to visit, and in what order?

1 WHAT TO VISIT



2 IN WHAT ORDER



Existing methods trade off personalization and feasibility

1 ROUTE OPTIMIZATION

OP • TTD

STRENGTH

Constraints by construction

LIMIT

Simple utility; weak personalization

2 PERSONALIZED + CONTEXT

CF • hybrid • CARS

STRENGTH

Learns user and context
relevance

LIMIT

Usually ranks POIs, not routes

3 DEEP SEQUENTIAL MODELS

RNN • Transformer • GNN

STRENGTH

Learns mobility transitions

LIMIT

Local objective; greedy route

RESEARCH GAP

Next-POI accuracy $\neq$ itinerary quality - and evaluation remains fragmented.

This thesis connects learned relevance to constrained route generation, then evaluates each task separately.

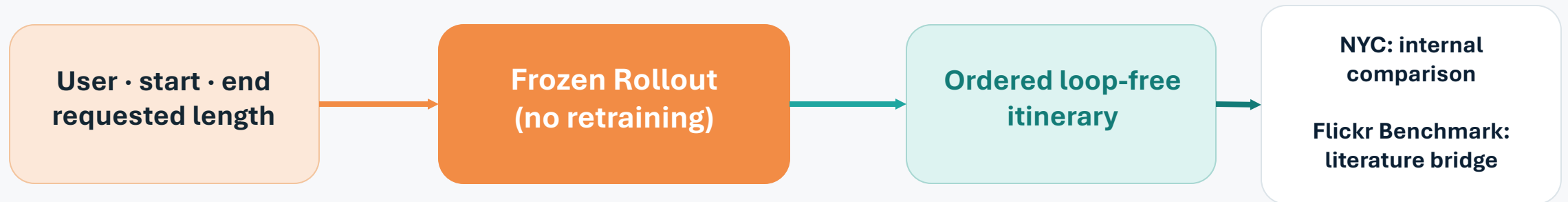
Source: Thesis Chapter 2, Sections 2.3-2.8; especially Table 2.3 and Research Gaps 3 and 5.

One engine, one decoder, and two fair evaluations

PHASE 1 · NEXT-POI



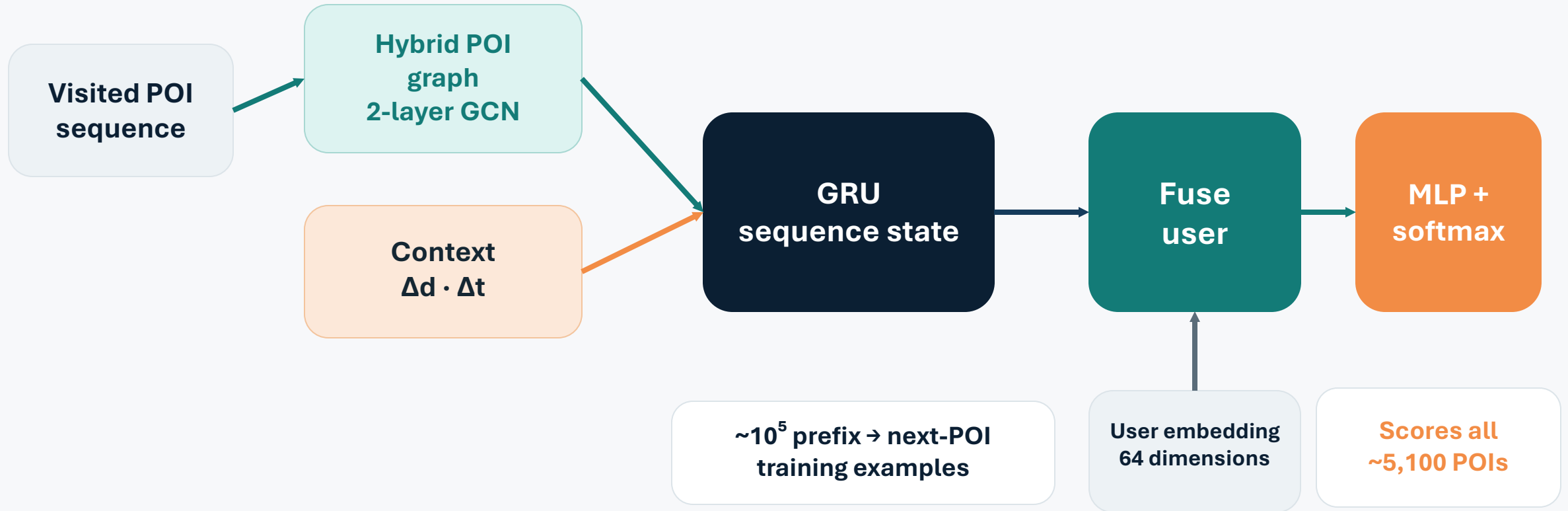
PHASE 2 · ITINERARY



Never mix scores across tasks, datasets, protocols, or metrics

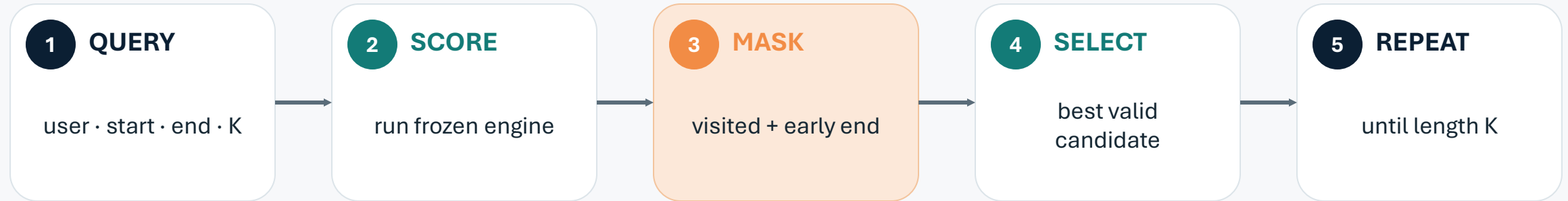
Source: Thesis Sections 3.2 and 3.6. Two tasks are evaluated separately.

The engine combines places, context, and user preferences



Source: Thesis Sections 3.3-3.4. Implemented context is distance and elapsed time.

Frozen Rollout turns next-POI scores into a loop-free route



The route grows one valid stop at a time



Source: Thesis Section 3.5. Decode-time constraints: loop-free, fixed endpoints, requested length.

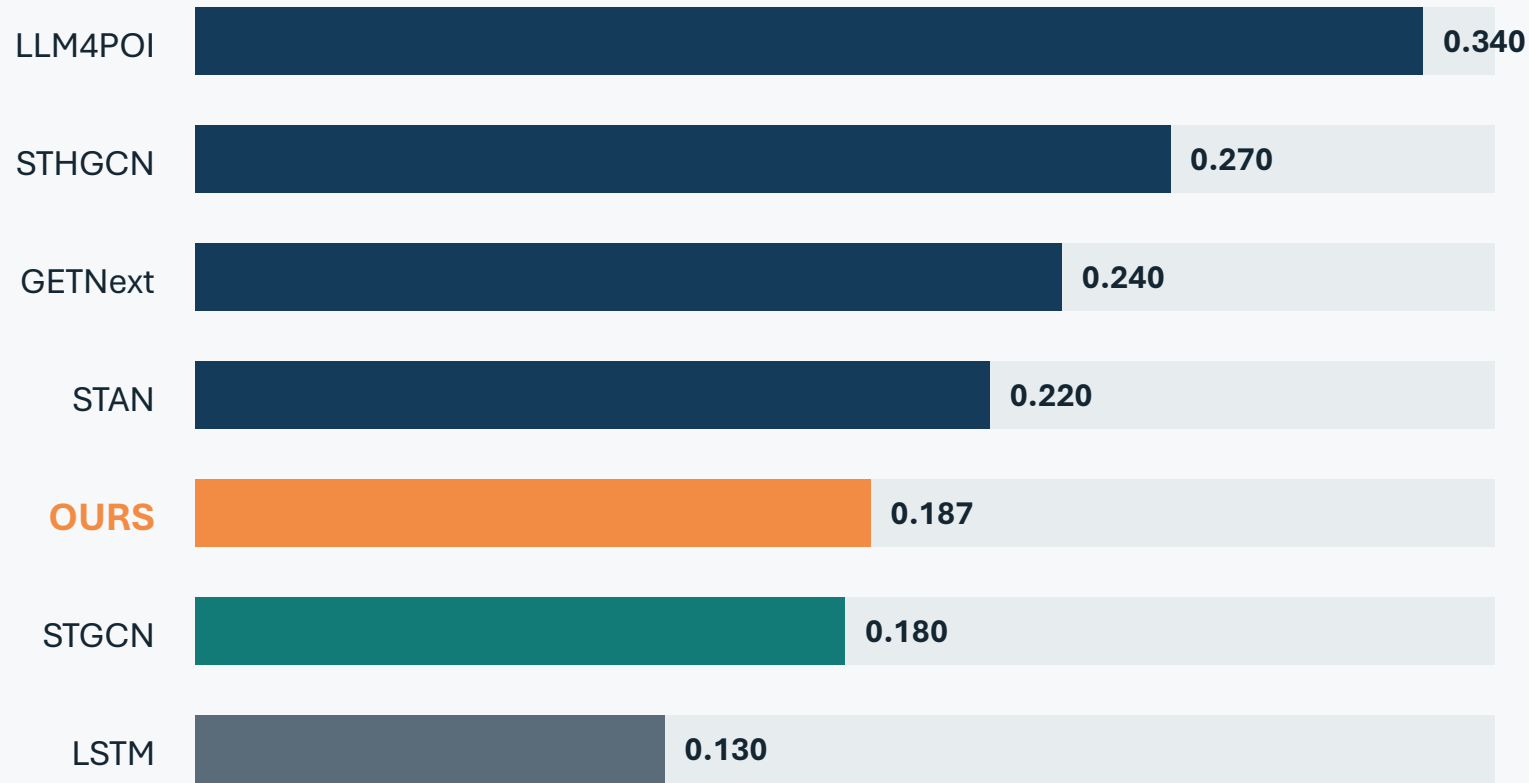
The engine is competitive, but not state of the art

0.187

HR@1

Exact next POI ranked first
~1 time in 5 among
~5,100

HR@10 = 0.588



Higher is better · full vocabulary · Foursquare NYC

Source: Thesis Table 3.2 and Figure 3.2. Published comparison values use the same NYC full-vocabulary HR@1 protocol.

Frozen Rollout wins; dense supervision helps explain why

FROZEN ROLLOUT

$\sim 10^5$

prefix → next-POI examples

Reuses a converged engine

0.289

pairs-F1

>

TRAINED POINTER

1

whole trajectory per session

Trains end-to-end from scratch

0.259

pairs-F1

NYC · n = 2,880 · length ≥ 3 · order-aware pairs-F1

Source: Thesis Table 3.3. Beam search changes Frozen Rollout only from 0.289 to 0.290.

Flickr validates the protocol - and shows the remaining gap

FLICKR BENCHMARK

27 - 88 POIs per city

PROTOCOL CHECK

Leave one trajectory out
First + last POI + length
Length ≥ 3 · loop-free · pairs-F1

0.021

maximum Random reproduction gap

Protocol faithfully reproduced

PAIRS-F1 ON THE PUBLISHED SCALE

Classical baselines

0.23 - 0.59

Trained Pointer

0.31 - 0.49

SelfTrip / AR-Trip

0.78 - 0.85

Our reproduced baselines reach the published scale; neural SOTA remains clearly ahead.

SEPARATE USER-EMBEDDING ABLATION

+0.038

Glasgow

-0.007

Osaka

-0.017

Toronto

Sparse leave-one-out history
creates cold-start.

Source: Thesis Tables 3.4-3.8 and Figures 3.4-3.5. NYC and Flickr scores are never compared directly.

Three results to remember

0.187 HR@1

0.289 > 0.259

max gap 0.021



honest next-POI engine
~5,100 NYC candidates

Frozen Rollout beats
the Trained Pointer

Flickr Benchmark reproduces
Chen 2016 Random

Compare only when dataset + protocol + metric are identical

Data preparation and evaluation protocols

PHASE 1 · FOURSQUARE NYC

- 1 **Raw data** ~227K check-ins · ~1.08K users · ~38K venues
- 2 **Filter** Iterative user/POI count ≥ 10
- 3 **Final catalog** ~5,100 POIs · ~1,000 users
- 4 **Sessions** Split after >24 h silence
- 5 **Split** Chronological per user: 70 / 10 / 20
- 6 **Evaluation** Full vocabulary · no sampled negatives

PHASE 2 · FLICKR BENCHMARK

City	POIs	Users	Traj.	Eval. ≥ 3
Toronto	29	1,395	6,057	335
Osaka	27	450	1,115	47
Glasgow	27	601	2,227	112
Edinburgh	28	1,454	5,028	634
Melbourne	88	1,000	5,106	442

Chen-2016 protocol

Leave one trajectory out · first + last POI + length given
length ≥ 3 · loop-free · pairs-F1

Metric definitions and complete result tables

NEXT-POI · NYC

Metric	Score
HR@1	0.187
HR@5	0.476
HR@10	0.588
NDCG@5	0.339
NDCG@10	0.375
MRR	0.316

FLICKR BENCHMARK · PAIRS-F1

Method	Toronto	Osaka	Glasgow	Edin.	Melb.
Random	.298	.301	.301	.270	.227
PoiPopularity	.443	.413	.510	.439	.320
Markov	.504	.421	.587	.449	.333
MarkovPath	.528	.398	.543	.452	.346
Trained Pointer · beam 3	.431	.399	.489	.414	.312

ITINERARY · NYC

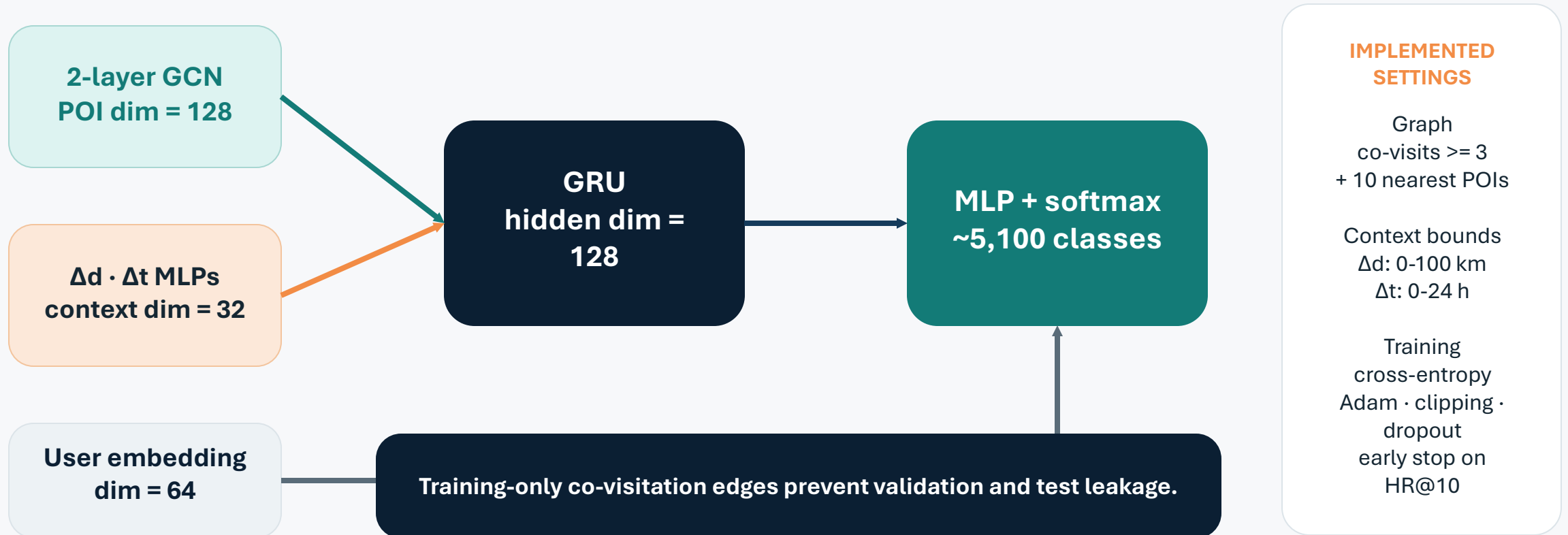
Method	pairs	set	exact
Frozen Rollout · greedy	.289	.609	.054
Frozen Rollout · beam 3	.290	.610	.057
Trained Pointer · v1	.259	.578	.043
Trained Pointer · v2 +ctx	.261	-	-

Source: Thesis Tables 3.2, 3.3 and 3.6. Dashes mean unreported, not zero.

METRIC BOUNDARY

HR@k / NDCG / MRR rank one next POI.
set-F1 measures recovered places. pairs-F1 measures order.
These values never share one leaderboard.

Detailed architecture, dimensions, and training controls



Source: Thesis Sections 3.3-3.4. Co-visitation edges are computed from training data only.

Why the Trained Pointer loses - limitations and next steps

TRAINING CONTROLS

0.259 → 0.261

Adding context recovers only +0.002

- ✓ validation pairs-F1 reaches a real peak
- ✓ early stopping applied
- ✓ context-aware variant tested
- ✓ Trained Pointer also trails Markov on the Flickr Benchmark

Conclusion: a genuine negative result for this tested setup, not a universal claim about all itinerary models.

MAIN LIMITATIONS

- below neural / LLM SOTA
- scorer remains locally myopic
- context only uses Δd and Δt
- decode-time Δt is assumed
- no opening hours or time windows
- one next-POI city
- offline evaluation only
- no deployment or user study

NEXT STEPS

- weather and time-of-day context
- persona/content cold-start encoder
- travel cost and opening hours
- schedule-aware constrained decoding
- pre-training and trajectory augmentation
- multi-city validation
- user study

Orienteering / ILS: scoped, NOT implemented

Source: Thesis Sections 3.7.5 and General Conclusion. Reader-facing method names are used throughout.